

When the Fed Tightens, Who in India Pays?

Moirai: US monetary policy through central banks, Indian banks and
200,000 heterogeneous households

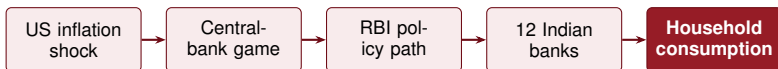
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When the Fed Tightens, Who in India Pays?

- ▶ **Question:** how does a US monetary tightening travel through international central banks → the RBI → Indian banks → **heterogeneous Indian households**?
- ▶ **Approach:** split the transmission into links that are **empirically checked** and links that are **explicitly assumed**, and declare which is which.



Headline finding

Per **100bp** of Fed tightening, floating-rate borrowers cut consumption **0.18 pp** more than savers in the static game and **0.55 pp** more in the dynamic game.

How the Shock Travels Through the Model



- ▶ **Data:** FRED/ALFRED and RBI sources; every series keeps its vintage and provenance.
- ▶ **Macro:** a US VAR identifies the *shape* of the response to a monetary-policy shock.
- ▶ **Strategic:** a game between five central banks determines the *size* of each bank's response to a US inflation shock.
- ▶ **Indian:** the RBI's path passes through heterogeneous commercial banks to 200,000 households.

The following slides zoom into each block in turn.

A Reproducible Information Set Comes First

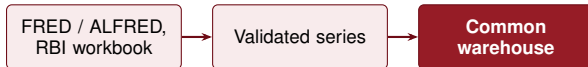
- ▶ Every series is **unit-labelled**, **bitemporal** (when a fact was true *and* when it was known) so revisions are traced, not silently overwritten.

US benchmark (FRED)

INDPRO	Industrial production (output)
CPIAUCSL	Consumer price index
FEDFUNDS	Effective federal funds rate

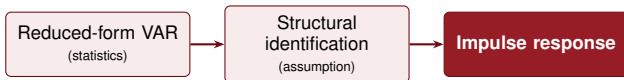
Indian evidence (RBI)

IIP	Output growth (y/y)
CPI	CPI inflation
Repo	Policy rate
G-Sec	10-year yield

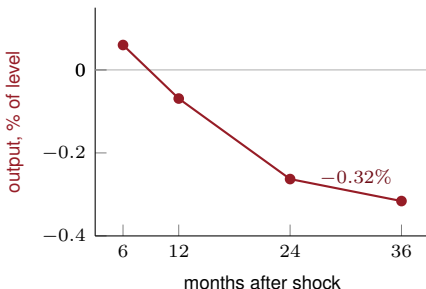


Does a Monetary Tightening Lower Output?

- ▶ **Specification:** VAR(12) in output growth, CPI growth and the change in the funds rate, 1985–2007, 269 monthly observations.
 - ▶ Passes the diagnostics gate: no residual autocorrelation; stable (max modulus 0.947); 6.9 observations per parameter.
 - ▶ The full 1960–2019 sample fails: it spans several policy regimes.
- ▶ **Identification:** recursive (Cholesky) ordering **output** → **prices** → **funds rate** the Fed sees this month's output and prices; they respond to it only with a lag. All six orderings are checked.
- ▶ **Role in the chain:** the VAR supplies the **shape** of the policy response, not the India-specific size of the shock.



The Policy Shock Lowers Output With a Lag



Cumulative response to a one-SD tightening, selected horizons

- ▶ Funds rate: about **+40bp** at the cumulative peak (months 10–12), still **+29bp** after three years.
 - ▶ Output: slightly up at first, then **−0.32%** at 36 months the classic lag.
 - ▶ The shock explains **under 6%** of output and price variation at 36 months.
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- ▶ **Robustness:** all six orderings keep the output sign; a proxy SVAR using high-frequency Fed surprises ($F = 18$) gives a similar household spread (**0.57** vs 0.61).
 - ▶ Prices rise (+0.14%) the price puzzle. It survives the proxy SVAR, and inflation is not transplanted to India.

Why Game Theory: Central Banks Are Strategic

- ▶ **Players:** Fed, ECB, BoJ, BoE, RBI, each setting its policy rate given domestic objectives and foreign spillovers.
- ▶ **Preferences:** each bank minimises a quadratic loss

$$L = w_{\pi}(\pi - \pi^*)^2 + w_y \text{gap}^2 + w_e (\text{external gap})^2 + w_s (r - r_{\text{current}})^2$$

- ▶ Published **mandates** discipline the weights: the Fed is dual; the ECB and BoJ put price stability first; the BoE and RBI are hierarchical (RBI: 4% CPI target, 2–6% band).
- ▶ The most consequential weight the Fed's output weight is **estimated** from its decisions.
- ▶ **Why not just regress?** A VAR describes historical co-movement. It cannot say what a central bank would do under conditions it has never faced; a game **derives behaviour from stated objectives**.

Trade-off, stated openly: the game answers counterfactuals a VAR cannot, but its payoffs are assumed preferences so every weight carries a confidence flag and a sensitivity check.

Nash, Stackelberg and the RBI's Best Response

- **Nash:** quadratic losses give linear first-order conditions; stacking all five gives $Ar = b$, solved exactly. No bank wants to move its rate alone.
- **Stackelberg:** the Fed leads, optimising along the other four's reaction function. Leadership changes rates by **under 1bp**.
- **RBI best response** per point of the Fed's rate (2013 validation):

Exchange (inflation)	+0.165
Demand (output)	−0.060
External differential	+0.023
Total	+0.128

Headline game (caused moves)

Bank	Total	Unprompted	Caused
Fed	+146	+58	+88
ECB	+26	+17	+9
BoJ	+34	+26	+8
BoE	+92	+79	+13
RBI	+19	+6	+12.9

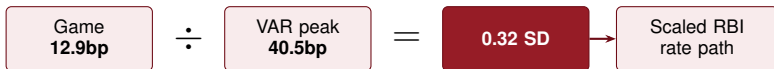
basis points; literature-anchored spillovers

RBI caused move
+12.9bp

Only **caused** moves are imported: the game is solved with and without the US condition, so what a bank would do anyway (RBI +6bp) is excluded.

From the Game's RBI Move to an Indian Path

- ▶ **Size** comes from the game: RBI caused move = **12.9bp**.
- ▶ **Shape** comes from the VAR: a one-SD shock peaks at **40.5bp** (cumulative).
- ▶ **Scaling**: the VAR response is rescaled to the game's move.

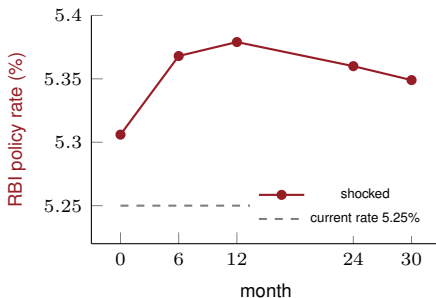


Declared departures (carried with the result, each with a reason):

- ▶ **Inflation held at baseline** the US price response contradicts Indian evidence.
- ▶ **Horizon capped at 30 months** the top of the Indian 8–10 quarter transmission range; US persistence is too long.

An explicit modelling seam not a direct empirical estimate of India's full macro response.

The Imported Tightening Is Gradual and Persistent



- ▶ Households are driven by a path **generated by the upstream chain**, not an arbitrary direct rate shock.
- ▶ The earlier headline the Fed's path at 2 SD sent straight to households is kept only as an **upper bound**.

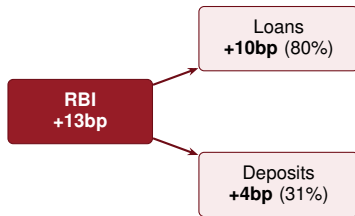
- ▶ RBI path: **5.306%** → **5.379%** at month 12 → **5.349%** at month 30.
- ▶ +5.6bp at once, +12.9bp by months 10–12, still +9.9bp at month 30.
- ▶ Inflation held at **4.4%**; income growth dips from 6.50% to 6.41% at month 12, then recovers.

What Reaches Borrowers and Savers?

12 banks in the RBI's three groups

Group	Credit share	Lending PT
Public (6)	58.8%	75%
Private (4)	38.8%	88%
Foreign (2)	2.4%	92%

PT = pass-through in tightening



- ▶ Public banks hold most credit and transmit least, pulling the system towards the weakest group.
- ▶ **Lending responds far more than deposits**: a borrower–saver wedge, with the difference kept by banks as margin.
- ▶ Held-out test (May 2022 – Nov 2024): lending **passes** (model 80% vs observed 76%).

Caveat: deposit pass-through **fails** its held-out test (31% modelled vs 97% observed) the model has no liquidity state.

Lending, Not Deposits, Drives the Spread

- ▶ Floating-rate borrowers face higher debt service because **loan rates respond strongly** and the lending side passed its held-out test.
- ▶ Stress test: rerun the pipeline with deposit pass-through at the **observed 97%**.

	Modelled deposits (31%)	Observed deposits (97%)
Aggregate consumption	−0.085%	−0.024%
Net savers	lose	gain (sign flips)
Borrower–saver spread	−0.61 pp	−0.60 pp

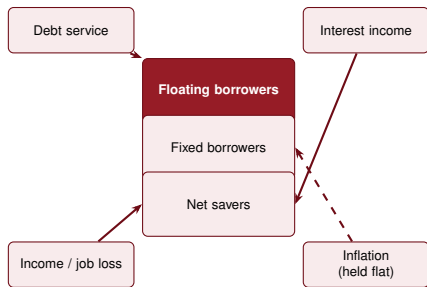
original household measure

Defensible conclusion

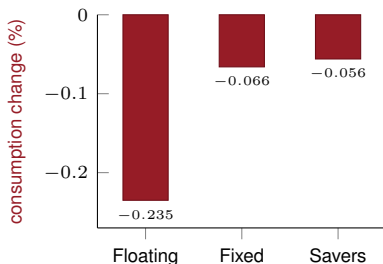
Deposit uncertainty moves the *aggregate* and the savers, but barely the *spread*: the distributional result rests on lending. That is why the spread is the headline and the aggregate is reported only as a range.

200,000 Households Turn Rates Into Spending

- ▶ Households differ in **income**, **liquid wealth**, **debt** and **employment**.
- ▶ Fitted to **7 Indian targets**; 7/7 once debt is capped at the lenders' limit ($4.22 \times$ income).
- ▶ Consumption = $MPC \times$ disposable income; the MPC falls with liquid wealth, from **0.70** (assumed) to **0.45** (sourced: Fagereng, Holm & Natvik 2021).
- ▶ Worked example: +10bp on loans $\Rightarrow$ **+Rs. 686** a year for the typical exposed household, i.e. 0.22% of income diverted to debt service.
- ▶ Deliberately **transparent** rather than a full estimated heterogeneous-agent model: every response decomposes into named channels.



Floating-Rate Borrowers Bear the Tightening



Change in each group's total consumption

Measuring a group's response

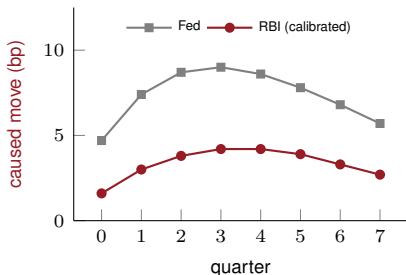
Population	Avg. of changes	Group totals
Uncapped	-0.61 pp	-0.18 pp
Debt capped	-0.20 pp	-0.16 pp

Borrower-saver spread

- ▶ The average of changes is inflated by **1,442** households near zero consumption.
- ▶ Capping debt removes them, and the two measures **converge**.
- ▶ **678** additional job losses over the horizon.

Income quintile	1	2	3	4	5
Consumption	-0.038%	-0.091%	-0.173%	-0.237%	-0.199%
% rate-exposed	3.0%	9.7%	20.5%	27.0%	21.4%

Dynamic Game: Timing and Follow-Through



- ▶ Quarterly choices linked by lags and **inertia**, calibrated only on the Fed's published smoothing (0.79).
- ▶ **Timing, out of sample:** the RBI path builds and persists like India's measured repo response shape gap **2.99** → **0.04**, and 0.031 after the final calibration.
- ▶ **Size:** the measured RBI response to US policy is **at least 0.47** per point; the RBI is calibrated to that floor.

Key interpretation

The static game's RBI follows the Fed by about **0.15**; the calibrated dynamic RBI by **0.47** roughly **three times** as closely. Since 0.47 is a floor, the dynamic consumption result is a **lower bound**.

Small in Aggregate, Large in Distribution



- ▶ **Main result:** per 100bp of Fed tightening, floating-rate borrowers cut consumption **0.18 pp** (static) to **0.55 pp** (dynamic, lower bound) more than savers.
- ▶ **Why the gap:** the dynamic RBI follows the Fed about three times as closely, so more of the Fed's move reaches floating-rate debt.
- ▶ **Interpretation:** the aggregate barely moves because the shock is largely a **transfer**; the burden falls on rate-exposed, especially **middle-income**, floating-rate borrowers.

Honest limitations

Household responses are calibrated, not validated against observed Indian spending (that needs a panel such as CMIE's CPHS); informal credit and household-specific inflation are not modelled; deposit pass-through failed its held-out test.

Conclusion

- 1 **The Fed's tightening reaches India through the RBI.** In the static game an 88bp Fed move causes a **+12.9bp** RBI move; measured data show the RBI follows the Fed by **at least 0.47** per point.
- 2 **Lending carries the transmission.** A 13bp RBI move reaches borrowers as **10bp** but savers as only **4bp**; lending passes its held-out test.
- 3 **The burden is distributional, not aggregate.** Per 100bp of Fed tightening, floating-rate borrowers cut consumption **0.18–0.55 pp** more than savers; the burden peaks among **middle-income** borrowers.
- 4 **Each link is checked where data allow:** VAR diagnostics, six orderings and a proxy SVAR; measured rupee and RBI responses; out-of-sample timing; a fresh-clone reproduction.

Next steps

Test the household layer against AIDIS and HCES; add informal credit and household-specific inflation; repair the deposit side with a liquidity state; pin data vintages for the historical tests.

References

US monetary transmission

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- ▶ Bernanke & Blinder (1992) funds rate as the policy indicator
- ▶ Sims (1992); Eichenbaum (1992) the price puzzle
- ▶ Kuttner (2001); Gürkaynak, Sack & Swanson (2005) high-frequency surprises
- ▶ Stock & Watson (2012, 2018); Mertens & Ravn (2013) external-instrument (proxy) SVAR
- ▶ Nakamura & Steinsson (2018) the Fed information effect
- ▶ Bauer & Swanson Fed policy surprises (San Francisco Fed)

Strategic transmission

- ▶ Clarida, Galí & Gertler (2000) Fed policy smoothing (0.79)
- ▶ Rey (2015) the global financial cycle

India

- ▶ Lakdawala (2021) India's sensitivity to US monetary policy
- ▶ Gopinath (2015) dollar invoicing of India's imports
- ▶ IMF Working Paper 2023/107 US tightening and EM bond yields
- ▶ RBI, *Monetary Policy Report*, October 2022 exchange-rate pass-through to CPI
- ▶ RBI, *Report on Currency and Finance* the external objective
- ▶ Patel Committee (2014) the CPI Combined target

Households

- ▶ Fagereng, Holm & Natvik (2021) MPC of liquid households (0.45)

Data

- ▶ FRED / ALFRED, Federal Reserve Bank of St. Louis
- ▶ RBI, Select Economic Indicators workbook